MACHINE LEARNING

2026-2029 ODD SEM

Project Problem Statement Submission Form

Sec No- 1	**Team No- 26**	**Project Title-** AUTOMATIC TEXT SUMMERISER
Team Lead Roll Number		2520030102
Team Lead Name		TARIGOPULA HASINI
Team Member Roll Number		2520030522
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1. Abstract

The Automatic Text Summariser is an NLP-based application designed to generate concise and meaningful summaries from lengthy news articles. The project first implements extractive summarisation using the Text Rank algorithm, which identifies and ranks important sentences based on their similarity and relevance within the article. The system uses datasets such as CNN/DailyMail and In shorts to process real-world news content and generate summaries containing the most significant information while reducing the overall length of the text.

To improve the quality and naturalness of summaries, the project also implements abstractive summarisation using a pre-trained T5 model from Hugging Face. Unlike extractive summarisation, T5 generates new sentences by understanding the context and meaning of the original article. The outputs from TextRank and T5 are compared using evaluation metrics such as ROUGE to analyse their effectiveness. This project demonstrates the practical application of Natural Language Processing, text summarisation, transformer models, and model evaluation in automatically converting lengthy news articles into shorter, informative summaries.

2. Module Mapping

Module Name	**Assigned Roll Number(s)**	**Signature**
Data Preprocessing & Extractive Summarisation	2520030102	
Abstractive Summarisation & Evaluation	2520030522	

3. Module Details :

Data Preprocessing and Extractive Summarisation Module:
collecting and preprocessing the news article datasets such as CNN/DailyMail and Inshorts. This includes cleaning the text, splitting articles into sentences, and preparing the data for summarisation. The student also implements the extractive summarisation approach using the Text Rank algorithm and Sumy library to identify and select the most important sentences from the news article.

Abstractive Summarisation and Evaluation :
implementing the abstractive summarisation approach using the pre-trained T5 model from Hugging Face. The student generates concise summaries by understanding the context and rewriting the information in new sentences. The student also compares the Text Rank and T5 summaries using evaluation metrics such as ROUGE and integrates both approaches into the final application for displaying the summarisation results.

4. Frameworks / Platforms / Software Used:

Python – Main programming language used for developing the project.
Hugging Face Transformers – Used for implementing the pre-trained T5 model for abstractive summarisation.
Sumy – Used for implementing extractive summarisation using TextRank.
NLTK – Used for text preprocessing and sentence tokenisation.
PyTorch – Used for running the T5 deep learning model.
Pandas – Used for loading and processing the datasets.
NumPy – Used for numerical and data processing operations.
ROUGE – Used for evaluating and comparing the generated summaries.
Streamlit – Used to develop the interactive user interface.
Jupyter Notebook / Google Colab – Used for development, experimentation, and testing.
CNN/DailyMail – Dataset used for news articles and reference summaries.
Inshorts – Dataset used for short news articles and summaries.
Git & GitHub – Used for version control and project management.

Guide Name and Signature (remarks if any) ______________________________